6d Holomorphic Theory, E_3 Chiral Structure, and Quantum Groups: A Frontier Working Note for Volume III

Working note

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1 The Established Core

1.1 3d holomorphic CS → Kac–Moody (Costello–Gwilliam)

1.2 5d holomorphic CS → affine Yangian (Costello 2013)

1.3 Procházka–Rapčák: $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$

1.4 Drinfeld center passage (297 tests across 4 suites)

1.5 MO R-matrix comparison (72 tests)

1.6 Chiral coproduct spin-2 (23 tests)

1.7 E_n hierarchy from holomorphic CS (47+141 tests)

2 The Conjectural Programme

2.1 6d holomorphic theory → E_3 factorization

2.2 Quantum toroidal from E_3 projection

2.3 Chiral CE deformation $d_{\boldsymbol{h}}$

2.4 E_3 Koszul duality

2.5 Universal defect as chiral quantum group

2.6 $K3 \times E$ via factorization homology

3 The Frontier (Genuinely New)

3.1 Zamolodchikov tetrahedron equation for $U_{q,t}$

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3.3 Two-parameter R-matrix $R(u, v)$

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3.5 K3 double current algebra quantization

3.6 Perturbative $\int_{K3} \mathcal{F}$ and the Igusa cusp form

3.7 Categorical S-matrix: $E_3 \to \mathrm{Sp}_4(\mathbb{Z}) \to \Phi_{10}$

4 The Bottleneck: Chain-Level S^3-Framing for CY-A$_3$

4.1 The obstruction

4.2 Existing approaches

4.3 Can the 6d route bypass CY-A$_3$?

5 Open Problems (Ranked by Impact)

1. **Chain-level S^3-framing for CY-A$_3$.** Construct the functor Φ at $d = 3$. All of Vol III downstream of CY-A depends on this. Impact:

unlocks CY-B, CY-D, and the entire $K3 \times E$ programme.

2. **6d algebraic framework for compact manifolds.** Extend the CFG factorization homology from $\mathbb{C}^3$ to $K3 \times E$. Would provide an alternative to Problem 1. Requires non-Lagrangian algebraic QFT on compact complex 3-folds.

3. E_3 **Koszul duality (Conj ??).** Prove B_{E_3} has three commuting differentials and coproducts, and that $\mathcal{F}^!$ has inverted parameters. Would establish the parameter-inversion symmetry $U^!_{q,t} \simeq U_{q^{-1},t^{-1}}$.

4. **Zamolodchikov tetrahedron equation for $U_{q,t}$.** Construct the tetrahedron R-matrix and verify the 3-simplex equation computationally at charge 2. First genuine E_3 invariant beyond the YBE.

5. **Two-parameter R-matrix $R(u, v)$.** Construct and verify the parametric YBE for the E_3 R-matrix on Fock spaces. Relates MO stable envelopes to the quantum toroidal braiding.

6. **Spin-3 chiral coproduct.** Extend the spin-2 scaffold (23 tests) to spin 3. First test of nonlinear OPE in the coproduct. Discriminates shadow class L from class C.

7. $\int_{K3} \mathcal{F}$ **computation.** Compute the perturbative factorization homology over K3 and match to the Fourier–Jacobi expansion of Φ_{10}. Would confirm the link between E_3 factorization and Siegel modular forms.

8. **Categorical S-matrix and $\mathrm{Sp}_4(\mathbb{Z})$.** Construct the functor from the E_3 braided category to $\mathrm{Sp}_4(\mathbb{Z})$-representations and identify Φ_{10} as the partition function.

9. **Algebraic vs topological E_3 over $\mathbb{Z}$.** Determine whether the two E_3 structures (Deligne conjecture vs configuration space) agree integrally or only rationally. Torsion discrepancy would affect the quantum toroidal algebra at roots of unity.

10. **Quantum group realization CY-C.** Construct $C(\mathfrak{g}, q)$ as a chiral quantum group. Currently `\begin{conjecture}` only. Requires resolution of Problems 1 or 2 as input.